

Lesson 5: Exponents & Laws of Indices — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Simplify $x^6 \times x^4$.

2. Simplify $y^9 \div y^3$.

3. Simplify $(a^4)^3$.

4. Evaluate 7^0 .

5. Simplify $3x^2 \times 4x^5$.

6. Simplify $(2x^3)^4$.

Section 2 — Core practice

7. Evaluate $27^{(2/3)}$.

8. Evaluate $16^{(3/4)}$.

9. Write $1/x^4$ using a negative index.

10. Evaluate $4^{(-1/2)}$.

11. Simplify $(8x^6)^{(1/3)}$.

12. Simplify $15x^5y^3 \div 3x^2y$.

Section 3 — Challenge & exam-style

13. Simplify $(2x^2y)^3 \times x^{-4}$, giving your answer in the form $ax^n y^m$.

14. Evaluate $(25/9)^{-1/2}$.

15. Solve $2^x = 32$ by writing 32 as a power of 2.

16. Simplify $(a^3b^{-2})^2 \div (a^4b)$.